MUTUAL FUND ANALYTICS DASHBOARD

Capstone Project Report

Submitted To

Bluestock Fintech

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1. Introduction

The objective of this project is to analyze mutual fund data and derive insights related to investor behavior, fund performance, transaction trends, and state-wise investment patterns.

The project covers the complete analytics pipeline including data ingestion, cleaning, database creation, exploratory data analysis, performance analytics, advanced analytics, and dashboard development

2. Datasets Used

The project uses the following datasets:

- Fund Master
- NAV History
- AUM by Fund
- Monthly SIP
- Category Inflow
- Industry Folio
- Scheme Performance
- Investor Transactions
- Portfolio Holdings
- Benchmark Returns

3. Data Cleaning

Data cleaning was performed using Python and Pandas.

Steps:
- Missing value handling
- Duplicate removal
- Date conversion
- Data validation
- Creation of processed datasets

4. SQL Database

A SQLite database was created to store cleaned mutual fund datasets.

Database operations included:
- Table creation
- Data insertion
- Analytical SQL queries

5. Exploratory Data Analysis

EDA was performed to identify trends and patterns.

Analysis Included:
- State-wise investments
- Gender-wise distribution
- Transaction type analysis
- Top performing funds
- Investor demographics

6. Performance Analytics

Performance metrics analyzed:

- 1 Year Return
- 3 Year Return
- 5 Year Return
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

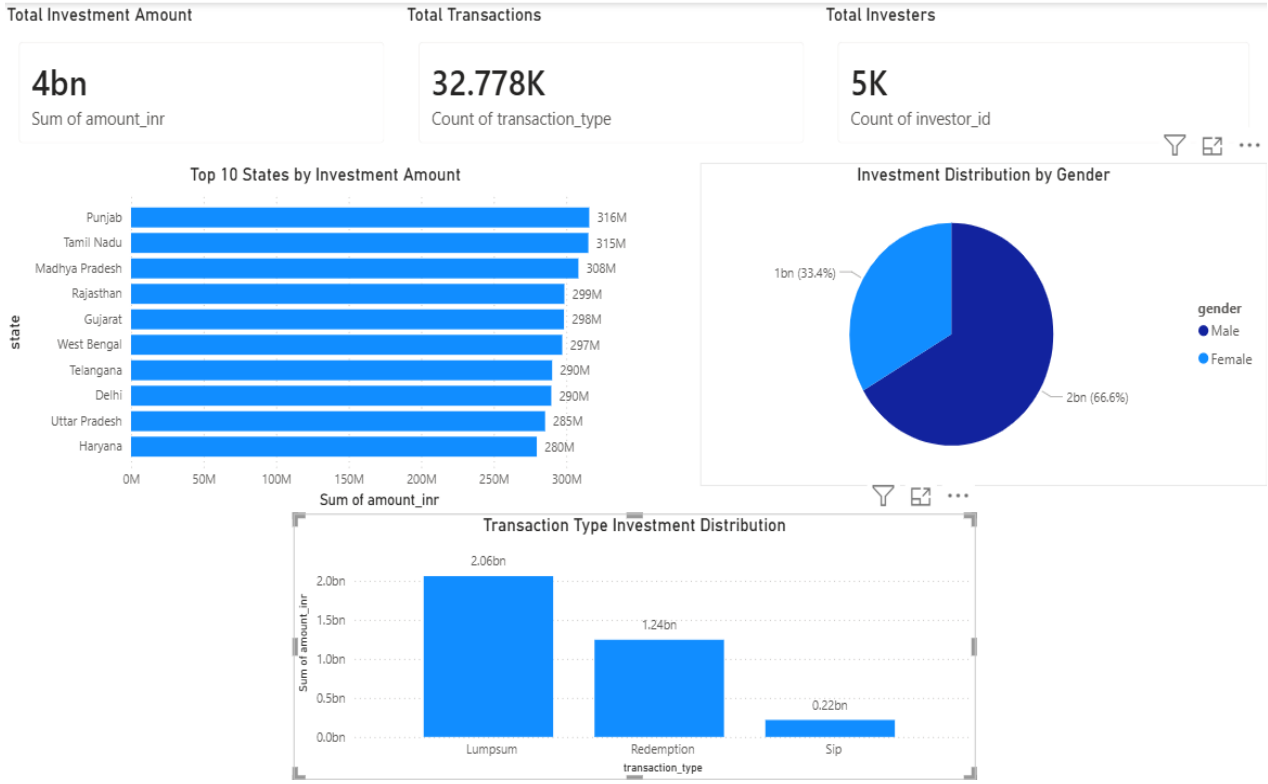
7. Advanced Analytics

Advanced analytics focused on:

- Investor behavior analysis
- Transaction insights
- Fund ranking
- Recommendation logic

8. Power BI Dashboard

Dashboard contains:
- Total Investment Amount
- Total Transactions
- Total Investors
- Top 10 States by Investment
- Transaction Type Distribution
- Gender-wise Distribution



9. Key Insights

1. Punjab and Tamil Nadu showed the highest investment amounts.

2. LumpSum investments contributed the largest investment volume.

3. Male investors contributed a higher investment amount compared to female investors.

4. Several schemes demonstrated strong long-term returns with favorable risk-adjusted performance.

10. Conclusion

The project successfully demonstrated an end-to-end mutual fund analytics workflow using Python, SQL, SQLite, and Power BI.

The developed dashboard and analytics pipeline can assist investors and analysts in understanding mutual fund trends and investment behavior.